

PG Management System – API Specifications

1. Overview

This document defines RESTful APIs for the PG Management SaaS platform. APIs are designed to be:

- Multi-tenant (org_id based)
- Subscription-aware
- Mobile & Web friendly
- Ready for Swagger / OpenAPI generation

Base URL

<https://api.pgmanagement.com/v1>

Common Headers

Header	Description
Authorization	Bearer
X-ORG-ID	Tenant / Organization ID
Content-Type	application/json

2. Authentication Module

2.1 User Login

POST /auth/login

Request

```
{
  "email": "owner@pg.com",
  "password": "*****"
}
```

Response

```
{
  "accessToken": "jwt-token",
  "refreshToken": "refresh-token",
  "user": {
    "id": "USR001",
    "role": "PG_OWNER",
    "org_id": "ORG001"
  }
}
```

```
}
```

3. Subscription Module

3.1 Create Subscription

POST /subscriptions

Request

```
{  
  "plan_id": "BASIC",  
  "billing_cycle": "MONTHLY"  
}
```

Response

```
{  
  "subscription_id": "SUB001",  
  "status": "ACTIVE",  
  "valid_till": "2026-01-31"  
}
```

3.2 Validate Subscription (Middleware API)

GET /subscriptions/validate

Response

```
{  
  "isActive": true,  
  "plan": "BASIC",  
  "features": ["MAP_LISTING", "BOOKINGS"]  
}
```

4. PG Management Module

4.1 Register PG (Owner First-Time Setup)

POST /pgs

Request

pg_management_api_specification

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```
{
  "pg_name": "Sri Lakshmi PG",
  "gender_type": "GENTS",
  "address": "T Nagar, Chennai",
  "location": {
    "latitude": 13.0418,
    "longitude": 80.2341
  }
}
```

Response

```
{
  "pg_id": "PG001",
  "status": "ACTIVE"
}
```

4.2 List PGs (Map & Search)

GET `/pgs/search?lat=13.04&lng=80.23&radius=5`

Response

```
[
  {
    "pg_id": "PG001",
    "pg_name": "Sri Lakshmi PG",
    "distance_km": 1.2,
    "available_beds": 4
  }
]
```

5. Room & Bed Module

5.1 Add Room

POST `/pgs/{pg_id}/rooms`

Request

```
{
  "room_type": "2_SHARING",
  "rent": 6500
}
```

5.2 Add Bed

POST /rooms/{room_id}/beds

6. Booking Module

6.1 Create Booking

POST /bookings

Request

```
{
  "pg_id": "PG001",
  "bed_id": "BED001",
  "move_in_date": "2026-02-01"
}
```

Response

```
{
  "booking_id": "BK001",
  "status": "CONFIRMED"
}
```

7. Payment & Billing Module

7.1 Generate Monthly Invoice

POST /billing/invoices

7.2 Pay Rent

POST /payments

8. Complaint Management

8.1 Raise Complaint

POST /complaints

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Request

```
{  
  "category": "ELECTRICITY",  
  "description": "Power issue in Room 101"  
}
```

8.2 Update Complaint Status

PUT `/complaints/{id}`

9. Error Handling (Common)

Code	Message
401	Unauthorized
403	Subscription Expired
404	Resource Not Found
422	Validation Error

10. Notes for Backend Team

- All APIs must enforce `org_id` isolation
- Geo queries must use PostGIS
- Subscription check is mandatory via middleware
- Ready for NestJS / Django REST implementation